

Néel-Vector-Dependent Altermagnetic Spin Splitting in the One-Dimensional Limit

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Altermagnetism is normally identified by nonrelativistic spin-split bands in a compensated collinear magnet. Under one-dimensional confinement, however, this diagnostic can disappear: a boundary-compatible sublattice-exchange operation may leave the only Bloch momentum unchanged, forcing the nonrelativistic spin-up and spin-down spectra to coincide. We show that nonrelativistic spin degeneracy can coexist with relativistic spin splitting in a compensated one-dimensional magnet. In a Lieb-based construction, confinement along the diagonal cancels the projected d -wave spin splitting while preserving the real-space sublattice-exchange motif. Spin-orbit coupling (SOC) then locks spin to the lattice, so the Néel-vector orientation selects a magnetic line group that can reveal spin-split bands. Using fully compensated [110] Ta₂TeSeO nanoribbons as a prototype, we find spin-degenerate nonrelativistic bands and sizable SOC splitting in the inherited easy-axis domain, $\mathbf{N} \parallel [100]$ or $[010]$. For sufficiently wide ribbons that retain the parent easy-axis order, this splitting is an equilibrium property. The easy-axis ribbon also supports a right-moving-mode spin polarization of about 35% at finite ideal conductance. Our results establish one-dimensional fully compensated altermagnetic functionality hidden inside apparently conventional antiferromagnetic bands.

The discovery of altermagnetism has shown that compensated collinear magnets can host spin-split bands without net magnetization [1–12]. In two dimensions, this idea has already led to both conventional altermagnets, whose nonrelativistic bands are spin split, and type-IV collinear magnets, whose nonrelativistic spectra remain spin degenerate but acquire relativistic time-reversal-symmetry-breaking responses under SOC [2, 3, 13–15]. These developments raise a sharper question at the ultimate limit of dimensional confinement. If an altermagnetic crystal is confined to one dimension along a symmetry-selected direction with compatible boundaries, the sublattice-exchange operation that produces momentum-dependent spin splitting in higher dimensions may act trivially on the only remaining Bloch momentum [1]. The usual band-structure fingerprint of altermagnetism can then disappear, making the system appear indistinguishable from an ordinary spin-degenerate antiferromagnet in the nonrelativistic spectrum.

The central issue is whether two confined magnets with equally spin-degenerate nonrelativistic bands must also have equivalent relativistic spin responses. The spatial operation exchanging opposite-spin sublattices can remain a symmetry of the confined ribbon even when its nonrelativistic bands are spin degenerate. This problem is complementary to general magnetic rod- and line-group classifications of one-dimensional spin splitting [16, 17]: instead of asking whether a one-dimensional magnetic group allows spin splitting in general, we ask how a boundary-enforced degeneracy inherited from an altermagnetic parent is converted into a relativistic channel response. With SOC, the symmetries compatible with the magnetic order depend on the Néel-vector orientation, allowing the nonrelativistic degeneracy to be lifted [18–22]. This distinguishes the proposed setting from a conventional $\mathcal{PT}$ -protected antiferromagnetic wire, where the same antiunitary symmetry would protect SOC double degeneracy

for any Néel direction. The Supplemental Material (SM) gives this case explicitly as a tight-binding control [23].

We therefore distinguish one-dimensional altermagnetic functionality from conventional nonrelativistic altermagnetic band splitting. The former is expressed through SOC-induced band splitting and spin-polarized longitudinal channels in a ribbon with a spin-degenerate nonrelativistic spectrum.

A minimal realization is a symmetric diagonal ribbon cut from a d -wave Lieb-type altermagnet [24–26]. In the two-dimensional parent, the sublattice-exchange operation swaps the two momentum components, so the nonrelativistic spin splitting is finite along the principal axes but vanishes on the diagonal. The [110] ribbon turns this operation into one that exchanges opposite-spin sublattices and the two edges while leaving the longitudinal momentum $k_{\parallel}$ unchanged. The ribbon can therefore look like an ordinary spin-degenerate antiferromagnet before SOC, although the real-space altermagnetic exchange structure remains. We demonstrate this scenario in ideal [110] Ta₂TeSeO nanoribbons, obtained from a Janus monolayer for which Néel-dependent band reconstruction and spin-polarized transport have been predicted [27]. Matched first-principles Wannier Hamiltonians with spin matrices yield spin-degenerate nonrelativistic ribbon bands and separated SOC branches with opposite Néel-axis spin projections for $\mathbf{N} \parallel [100]$. For sufficiently wide ribbons, the parent easy axes [100] and [010] make this spin-split domain the natural equilibrium candidate. The resulting ideal right-moving-mode polarization reaches approximately 35% at finite conducting-channel weight, while other Néel orientations give distinct spectral and channel responses. These results connect boundary-enforced nonrelativistic degeneracy to a sizable intrinsic one-dimensional altermagnetic spin response. The SM also gives a standalone CrCl₃ chain calculation, showing that the same diagnostic can be applied to a genuine one-dimensional structure not obtained by cutting a

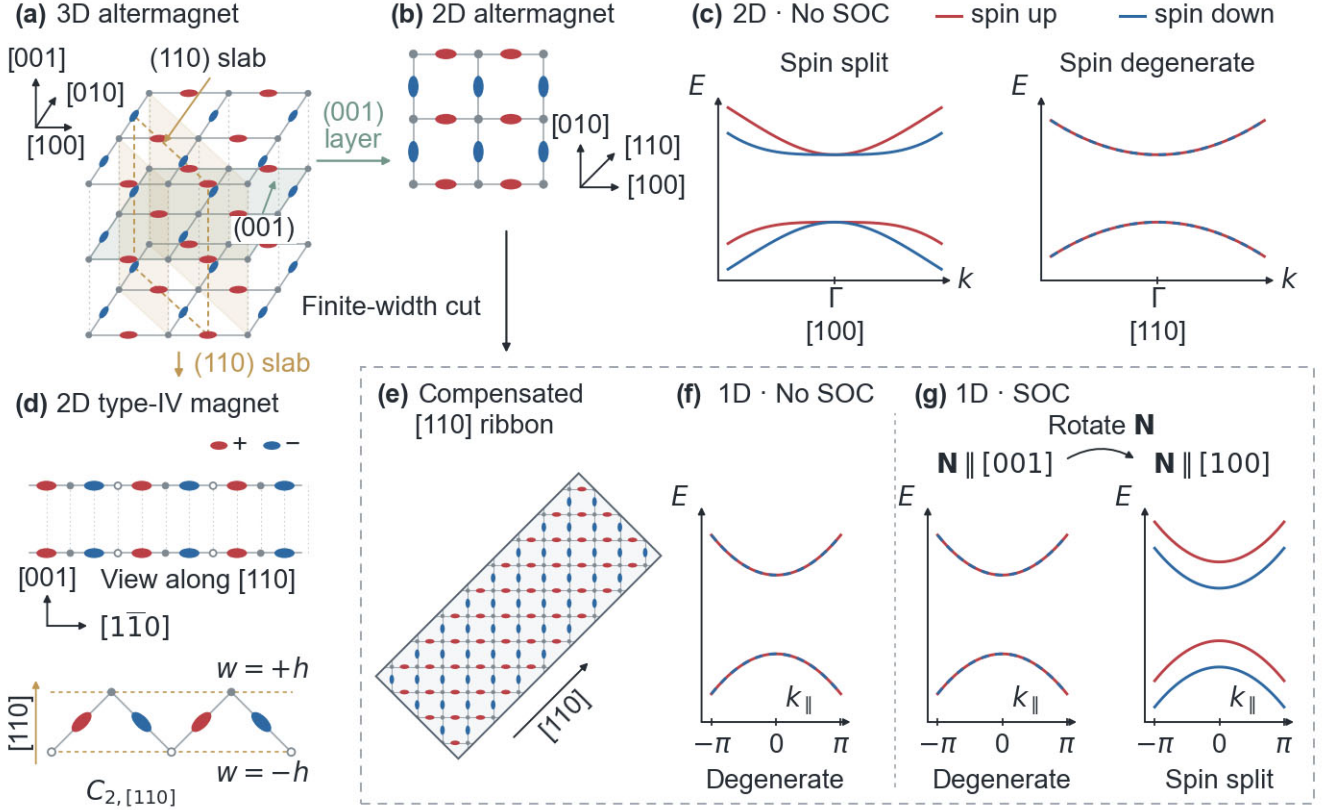


FIG. 1. Dimensional reduction and Néel-vector-controlled spin splitting in an ideal Lieb-based symmetry construction. (a) AA-stacked magnetic Lieb parent, with the selected (001) layer and (110) slab highlighted. (b) Two-dimensional Lieb altermagnet. (c) Nonrelativistic dispersions showing spin splitting along [100] and degeneracy along [110]. (d) (110) slab realizing the type-IV limit through the spin-exchanging $C_{2,[110]}$ symmetry; solid and open gray sites mark the two nonmagnetic sublayers. (e) Compensated [110] ribbon with symmetry-related edges. (f) Non-SOC ribbon bands remain spin degenerate. (g) SOC locks spin to the lattice, so rotating $\mathbf{N}$ from [001] to [100] can activate symmetry-allowed spin splitting. Red and blue denote opposite spin sectors or projected spin character; gray sites are nonmagnetic. All dispersions are schematic.

two-dimensional parent.

The dimensional construction in Fig. 1 begins with an ideal AA-stacked three-dimensional parent made of magnetic Lieb layers [Fig. 1(a)]. Selecting a single (001) layer gives a two-dimensional d -wave Lieb altermagnet [Fig. 1(b)]. In this layer, the sublattice-exchange operation swaps the two in-plane momentum components,

$$E_{\uparrow}(k_x, k_y) = E_{\downarrow}(k_y, k_x),$$

so the leading nonrelativistic splitting has the form $\Delta E \propto k_x^2 - k_y^2$. The spectrum is spin split along the principal directions [100] and [010], but remains degenerate along the diagonal [110] [Fig. 1(c)] [2, 28].

A finite (110) slab of the same parent gives the two-dimensional descendant shown in Fig. 1(d). The spin-exchanging $C_{2,[110]}$ operation acts within the slab Brillouin zone and produces the type-IV limit: the nonrelativistic spectrum is spin degenerate, while the real-space sublattice-exchange motif of the altermagnetic parent remains. The one-dimensional ribbon applies the same idea to the diagonal direction of the Lieb layer. A compensated ribbon periodic

along [110] has symmetry-related edges [Fig. 1(e)]. Let $b(\mathbf{r})$ be the collinear exchange field and let g be a spatial operation that maps the finite ribbon, including both edges, onto itself. If g reverses the exchange field while leaving the only Bloch momentum unchanged, then

$$b(g\mathbf{r}) = -b(\mathbf{r}), \quad gk_{\parallel} = k_{\parallel} \implies E_{\uparrow}(k_{\parallel}) = E_{\downarrow}(k_{\parallel}). \quad (1)$$

Without SOC, combining g with a π spin rotation about an axis perpendicular to $\mathbf{N}$ maps each spin-up eigenstate onto a spin-down eigenstate at the same $k_{\parallel}$ and energy. The finite ribbon is therefore spin degenerate throughout the one-dimensional Brillouin zone [Fig. 1(f)], even though it retains the real-space sublattice-exchange structure of the parent altermagnet.

With SOC, spin rotations are tied to the lattice, so a symmetry operation must preserve both the atomic arrangement and the magnetic moments [18]. An operation exchanging opposite-spin sublattices must also reverse the moments, either through its spatial action on spin or in combination with time reversal. Rotating the Néel vector can therefore preserve or remove these operations and change the ribbon's magnetic

line group.

An antiunitary symmetry that leaves $k_{\parallel}$ unchanged and squares to -1 protects an orthogonal equal-energy pair. SOC can lift the nonrelativistic degeneracy when the remaining symmetries no longer enforce this pairing. As sketched in Fig. 1(g), rotating $\mathbf{N}$ from $[001]$ to $[100]$ can convert a spin-degenerate ribbon spectrum into separated SOC branches in the ideal model. This is a relativistic response of the underlying compensated motif, not ordinary nonrelativistic spin splitting in a strictly one-dimensional Brillouin zone. If the easy axis selects such a domain, no external reorientation is needed to produce the splitting.

This mechanism also applies to one-dimensional structures whose nonrelativistic degeneracy satisfies Eq. (1) and is no longer enforced by the symmetries of the SOC magnetic domain. A higher-dimensional parent is therefore not required. Further symmetry analysis is given in the SM [23].

Ta_2TeSeO provides a material realization of this symmetry logic. The parent Janus monolayer contains a square Ta network decorated asymmetrically by Te, Se, and O. Its non-magnetic structure has space group $P4mm$ (No. 99) and point group C_{4v} , with the Janus polarity removing the horizontal mirror [27]. In the compensated collinear magnetic state, the nonrelativistic bands form an altermagnetic Weyl semimetal: spin splitting appears along the principal axes, whereas diagonal nodal directions remain degenerate. This is precisely the electronic structure needed for the Lieb-type construction above. Importantly, previous calculations identify $[100]$ and $[010]$ as the magnetic easy axes, not the diagonal $[110]$ direction chosen as the ribbon axis.

We therefore construct the prototype $[110]$ Ta_2TeSeO ribbon shown in Fig. 2(a), periodic along $[110]$ and finite along $[1\bar{1}0]$. The mirror-symmetric Ta-terminated ribbon spans 11.64 nm and contains 197 atoms per axial period, with 40 Ta sites in each magnetic sublattice. Its symmetric termination preserves the sublattice-exchange motif required by Eq. (1). The electronic structure is obtained by cutting the spin-resolved monolayer Wannier Hamiltonian into a finite strip, with computational details given in the SM [23].

Without SOC, the ribbon bands are spin degenerate [Fig. 2(b)]. The ribbon therefore keeps the compensated sublattice-exchange motif of the parent altermagnet, while $[110]$ confinement hides the familiar nonrelativistic spin splitting.

We then include SOC in the Wannier representation. The Hamiltonian and the physical spin operator σ are transformed into the same Wannier gauge, allowing the spin expectation value of each ribbon eigenstate to be evaluated directly. Benchmark comparisons with monolayer DFT spin-projected bands confirm that the Wannier model reproduces the low-energy dispersions and spin character. Details are given in the SM [23].

Figure 3 colors each eigenstate by its spin expectation value along the corresponding Néel vector,

$$s_N(n, k_{\parallel}) = \langle \psi_{nk_{\parallel}} | \sigma \cdot \hat{\mathbf{N}} | \psi_{nk_{\parallel}} \rangle, \quad (2)$$

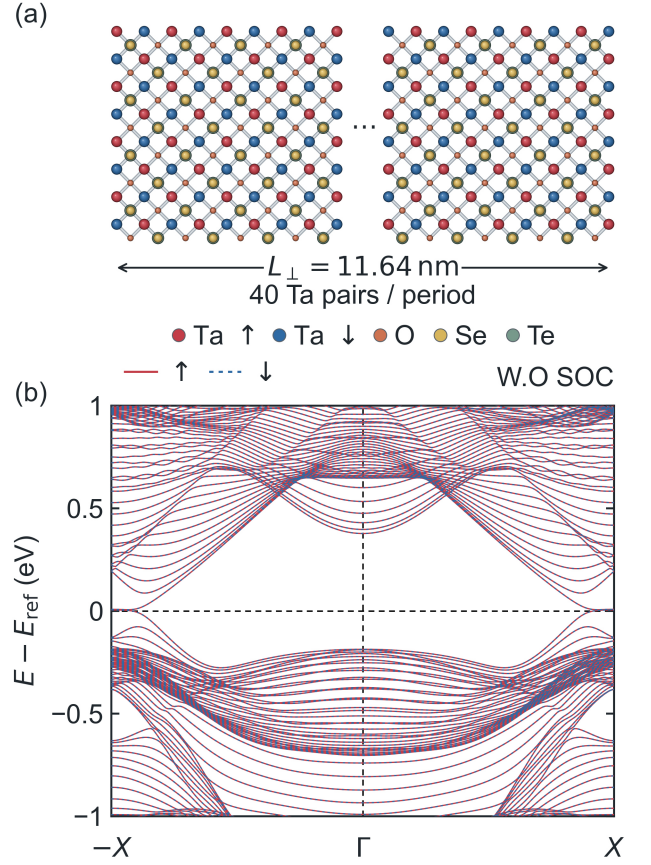


FIG. 2. Mirror-symmetric $[110]$ Ta_2TeSeO ribbon and non-SOC bands. (a) Ta-terminated ribbon with the central width omitted for clarity. (b) Wannier bands without SOC. Red solid and blue dashed curves are the two spin channels; black dashed lines mark E_{ref} and Γ along $-X-\Gamma-X$.

where σ is the spin Pauli vector. Red and blue denote positive and negative s_N for each magnetic domain, rather than projection onto a fixed ribbon axis. Because SOC mixes spin, the color scale represents spin expectation values, not conserved spin-channel occupations.

We use independent SOC Wannier Hamiltonians for $\mathbf{N} \parallel [100]$, $[001]$, and $[110]$. The $\mathbf{N} \parallel [1\bar{1}0]$ domain is obtained from the C_{4z} -related $[110]$ ribbon construction, with the Hamiltonian, spin operator, magnetic sublattices, and termination transformed consistently into the same final ribbon frame. This gives the symmetry-related spectrum shown in Fig. 3(c). Details are provided in the SM [23].

For the easy-axis domain $\mathbf{N} \parallel [100]$ [Fig. 3(a)], SOC produces clearly separated branches with opposite signs of s_N at the same momentum. The compensated ribbon thus exhibits a spin-split relativistic spectrum, in contrast to its non-SOC bands [Fig. 2(b)]. The transverse diagonal orientation $[1\bar{1}0]$ [Fig. 3(c)] also exhibits pronounced Néel-axis spin contrast.

For the $[001]$ and longitudinal $[110]$ domains [Figs. 3(b,d)], the low-energy states have much smaller projections along

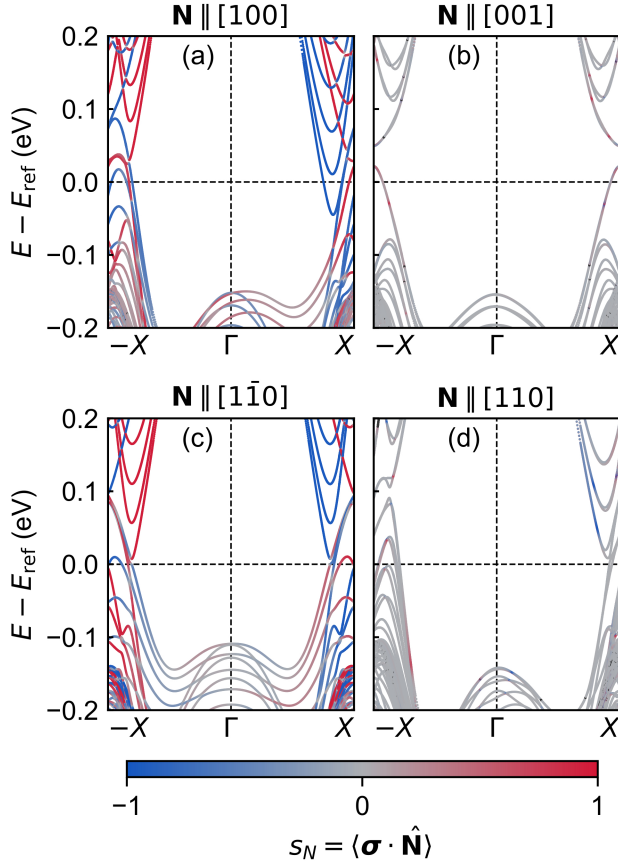


FIG. 3. SOC bands of the mirror-symmetric Ta-terminated [110] Ta_2TeSeO ribbon for $\mathbf{N} \parallel$ (a) [100], (b) [001], (c) $[1\bar{1}0]$, and (d) [110]. Red and blue denote positive and negative spin projection $s_N = \langle \boldsymbol{\sigma} \cdot \hat{\mathbf{N}} \rangle$ along the respective Néel axis. Light gray marks small projection, while dark gray marks states within 0.1 meV of another band. Panel (c) uses the C_{4z} -related construction. The path is $-X-\Gamma-X$. Black dashed lines mark $E - E_{\text{ref}} = 0$ and Γ , where E_{ref} is the reference of each magnetic domain.

their respective Néel axes over broad momentum ranges, although selected branches remain polarized.

The Ta_2TeSeO monolayer has in-plane [100]/[010] easy axes. For sufficiently wide [110] ribbons that retain the parent easy-axis order, the SOC-induced spin splitting in Fig. 3(a) is an equilibrium property rather than a response requiring external Néel-vector rotation.

The experimentally relevant question is how this intrinsic splitting is read out in transport. Figure 4(a) sketches a gated two-terminal device [6, 29]. We calculate the ideal polarization of right-moving modes with a fixed analysis axis $\hat{\mathbf{q}} \parallel [110]$, which compares the same physical spin component for all magnetic domains. This mode average is distinct from the band coloring, because oppositely polarized branches can cancel when all right-moving channels at a given energy are included.

For a unit-transmission benchmark, let $\mathcal{R}(E)$ denote the set of right-moving Bloch modes at energy E , namely states with

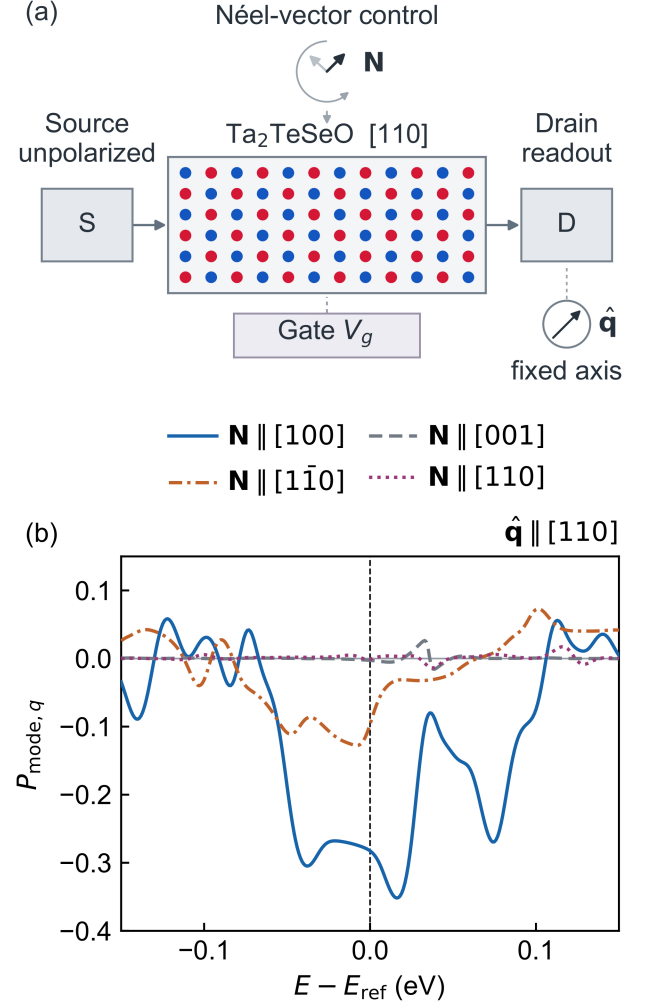


FIG. 4. Spin-polarized modes of the mirror-symmetric Ta-terminated [110] ribbon. (a) Gated two-terminal readout with fixed spin-analysis axis $\hat{\mathbf{q}} \parallel [110]$. (b) Right-moving-mode polarization $P_{\text{mode},q}(E)$ for $\mathbf{N} \parallel$ [100], [001], $[1\bar{1}0]$, and [110] in blue solid, gray dashed, orange dash-dotted, and purple dotted lines. Ideal conductance and numerical details are given in the SM.

positive group velocity along the ribbon, and let $N_+(E) = |\mathcal{R}(E)|$. For each mode v , $s_{vq}(E) = \langle \psi_v | \boldsymbol{\sigma} \cdot \hat{\mathbf{q}} | \psi_v \rangle$ is the spin expectation value along the fixed analysis axis $\hat{\mathbf{q}}$. We define

$$P_{\text{mode},q}(E) = \frac{1}{N_+(E)} \sum_{v \in \mathcal{R}(E)} s_{vq}(E), \quad G_{\text{ideal}}(E) = \frac{e^2}{h} N_+(E). \quad (3)$$

Here $P_{\text{mode},q}$ is the dimensionless right-moving-mode spin polarization, and G_{ideal} is the corresponding Landauer conductance for perfectly transmitting channels [30]. Because SOC removes a conserved up/down channel index, Eq. (3) is not the collinear ratio $(G_{\uparrow} - G_{\downarrow})/(G_{\uparrow} + G_{\downarrow})$. It is the unit-transmission limit of a Landauer spin-density-matrix readout along $\hat{\mathbf{q}}$ [31]. The plotted curves use 5 meV Gaussian broadening of the numerator and denominator separately. Contact-

dependent drain polarization and numerical details are discussed in the SM [23].

Figure 4(b) shows that the easy-axis [100] domain gives the dominant transport spin signal, reaching $P_{\text{mode},q} \simeq -0.352$ near $E - E_{\text{ref}} = 16$ meV with $G_{\text{ideal}} \simeq 4.47 e^2/h$. Thus the intrinsic band splitting produces sizable spin-polarized one-dimensional channels while several conducting modes remain available. The transverse-diagonal $[1\bar{1}0]$ domain reaches about 12.8%, whereas the [001] and longitudinal [110] domains remain at the percent level, about 1.9% and 1.7%, respectively. These small values refer only to the fixed $\hat{q}$ component. The [100] domain therefore provides the largest right-moving-mode polarization along the fixed detector axis, while the [001] and [110] domains yield much smaller signals.

In summary, compensated one-dimensional ribbons can exhibit SOC-induced spin splitting despite spin-degenerate nonrelativistic bands. We demonstrate this behavior in [110] Ta_2TeSeO nanoribbons, where the [100] domain exhibits separated branches with opposite Néel-axis spin projections and a right-moving-mode polarization of about 35% at finite ideal conductance. For wide ribbons that retain the parent easy-axis order, this spin-split state is an equilibrium property. The CrCl_3 chain provides an independent one-dimensional material example, whereas the $\mathcal{PT}$ -protected tight-binding wire retains its spin degeneracy with SOC. These contrasting cases show that the magnetic symmetry, rather than nonrelativistic spin degeneracy alone, determines whether a compensated one-dimensional magnet can support spin-split relativistic bands.

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